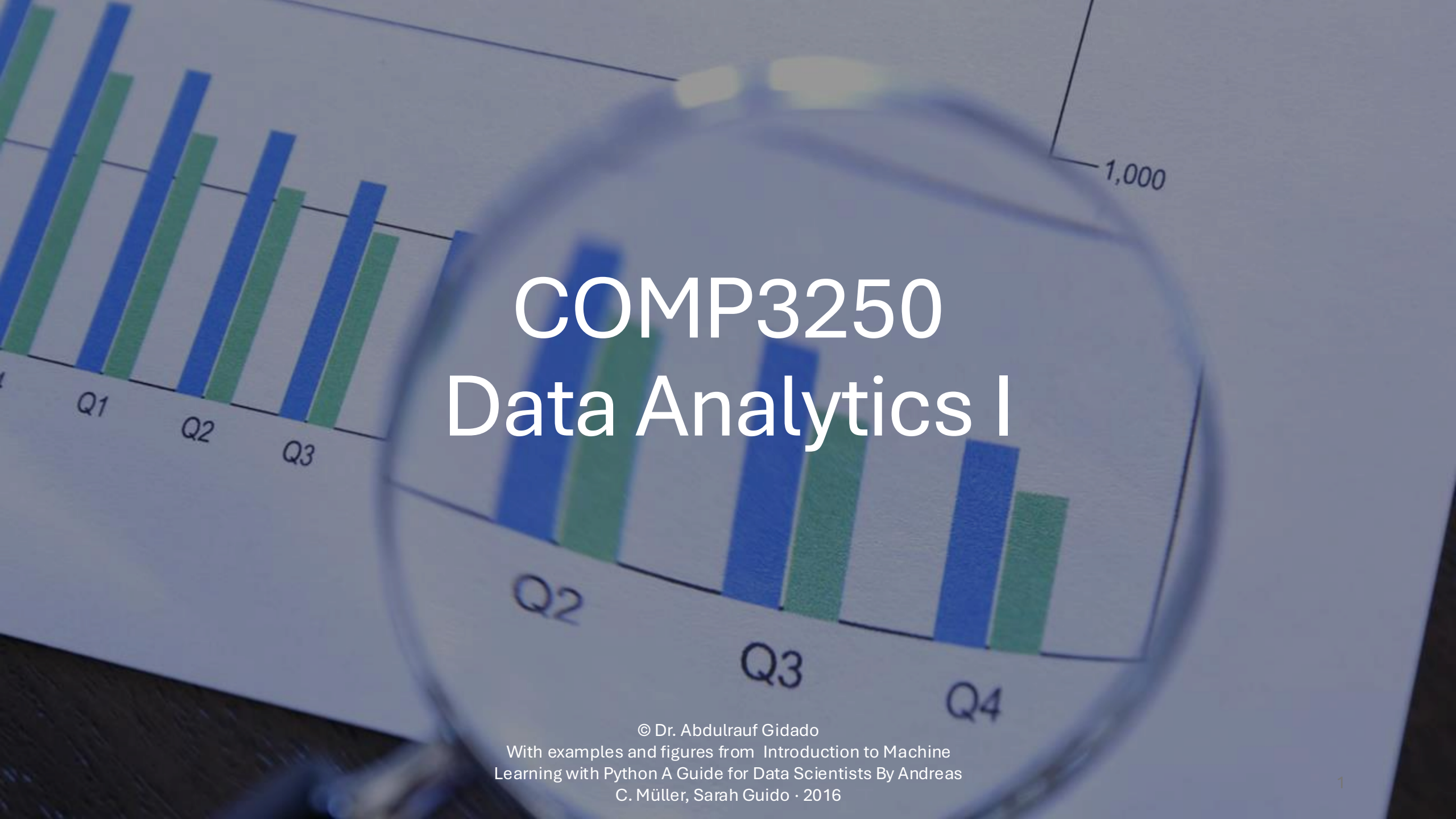




COMP3250

Data Analytics I

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With examples and figures from Introduction to Machine
Learning with Python A Guide for Data Scientists By Andreas
C. Müller, Sarah Guido · 2016

Agenda



Let's get to know
each other



Getting the best
out of the course:
Best practices



Course Outline



Introduction: Data
Analytics
Problems and ML.



Python Libraries
and tools for this
course



Exploring Data
and Models

Machine Learning



It is about **extracting knowledge** from **data**. It is a research field at the **intersection** of **statistics, artificial intelligence, and computer science** also



Also known as **predictive analytics** or **statistical learning**

Why Machine Learning?

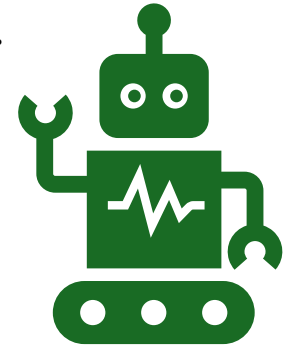


Intelligent
applications of then...

Spam: This would be an example of using an expert-designed rule system to design an “intelligent” application. Manually crafting decision rules is feasible for some applications, particularly those in which humans have a good understanding of the process to model

- *The logic required to decide is specific to a single domain and task. Changing the task even slightly might require a rewrite of the whole system.*
- *Designing rules requires a deep understanding of how a decision should be made by a human expert.*

Intelligent applications of now...



Supervised Machine Learning

- Machine learning algorithms that learn from input/output pairs are called supervised learning algorithms because a “teacher” provides supervision to the algorithms in the form of the desired outputs for each example that they learn from.
- **Identifying the zip code from handwritten digits on an envelope.** Here, the input is a scan of the handwriting, and the desired output is the actual digits in the zip code. To create a dataset for building a machine learning model, you need to collect many envelopes. Then, you can read the zip codes yourself and store the digits as your desired outcomes.
- **Determining whether a tumor is benign based on a medical image.** Here the input is the image, and the output is whether the tumor is benign. To create a dataset for building a model, you need a database of medical images. You also need an expert opinion, so a doctor needs to look at all of the images and decide which tumors are benign and which are not. It might even be necessary to do additional diagnosis beyond the content of the image to determine whether the tumor in the image is cancerous or not.

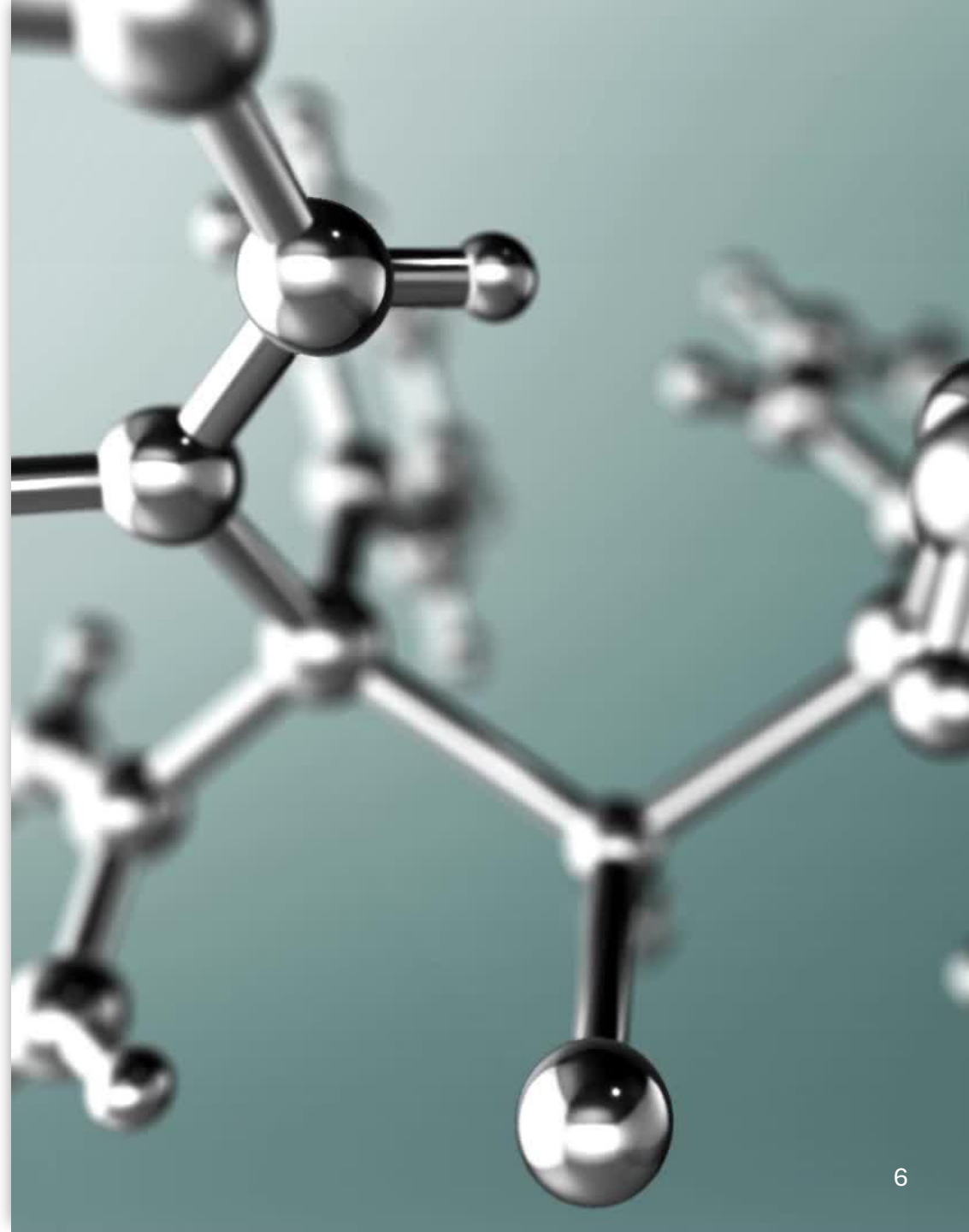
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Unsupervised ML Algorithms

- Only the input data is known, and no known output data is given to the algorithm. While there are many successful applications of these methods, they are usually harder to understand and evaluate.



Examples: Unsupervised Learning Problems

- Identifying topics in a set of blog posts
 - Segmenting customers into groups with similar preferences
 - Detecting abnormal access patterns to a website
-
- For both supervised and unsupervised learning tasks, it is important to have a representation
 - of your input data that a computer can understand. Often it is helpful to
 - think of your data as a table.

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Notes Dataset:

- Each data point that you want to reason about (each email, each customer, each transaction) is a row and each property that describes that data point (say, the customer's age or the amount or location of a transaction) is a column. You might describe users by their age, their gender, when they created an account, and how often they have bought from your online shop. You might describe the image of a tumor by the grayscale values of each pixel, or maybe by using the size, shape, and color of the tumor.

Notes on Dataset

- Each entity or row here is known as a **sample (or data point)** in machine learning, while the columns—the properties that describe these entities—are **called features**.

Further information on Datasets



One of the most important parts of the machine learning process is understanding the data you are working with and how it relates to the task you want to solve.



Each algorithm is different regarding what kind of data and what problem setting it works best for.

Always keep the following in mind while building a machine - learning solution:



What question(s) am I trying to answer? Do I think the data collected can answer that question?



What is the best way to phrase my question(s) as a machine - learning problem?



Have I collected enough data to represent the problem I want to solve?



What features of the data did I extract, and will these enable the right predictions?



How will I measure success in my application?



How will the machine learning solution interact with other parts of my research or business product?

Language, Libraries and Tools

- In this course, we primarily use Python Programming Language with the following libraries and tools for class demonstrations, assignments and midterms.
- ***scikit-learn***: It contains many state-of-the-art machine learning algorithms, as well as comprehensive documentation about each algorithm.
- **Ananconda**: A Python distribution made for large-scale data processing, predictive analytics, and scientific computing. Anaconda comes with NumPy, SciPy, matplotlib, pandas, IPython, Jupyter Notebook, and scikit-learn

Class Coding Demonstrations and Explanation of the Tools and Libraries

- Installing Anaconda
- Using Jupyter notebook
- Numpy
- SciPy
- matplotlib
- Pandas
- If you are new to Python and Anaconda, check the resources tab for guides on how to install them.
- *Ensure always to be present and attentive and participate in live coding class activities. It is important for us to have an impactful lesson and your success in this course!*